

The coordinate method in practice

Choose your axes well and a hard proof becomes two lines of arithmetic.

LESSON 38

1 × 40 MIN

GEOMETRY 8, TEMA 34

STAGE 9 · 3.1

LESSON CLOCK

5'

12'

12'

7'

4'

Place a rectangle on axes three ways

5 min

Proving with coordinates

12 min

Classifying triangles

12 min

Finding a missing vertex

7 min

Homework

4 min

LEARNING OBJECTIVES

- Place a figure on axes so that its coordinates are as simple as possible.
- Prove that a quadrilateral is a parallelogram, rectangle or rhombus using coordinates.
- Classify a triangle from the lengths of its sides.
- Find an unknown vertex from a condition.

TEXTBOOK

National

Tema 34, pp. 75–77

Cambridge

Learner’s Book pp. 56–60

Workbook

Workbook 3.1

01

Explanation

Choose the axes yourself

When a problem gives you a shape but no coordinates, **you** choose where to put it. A good choice puts one vertex at the origin and one side along an axis, so that as many coordinates as possible are zero.

SHAPE	A GOOD PLACING
Rectangle	$(0;0), (a;0), (a;b), (0;b)$
Right triangle	$(0;0), (a;0), (0;b)$
Isosceles triangle	$(-a;0), (a;0), (0;h)$
Parallelogram	$(0;0), (a;0), (a+b;c), (b;c)$

The letters keep the proof general — it then holds for **every** rectangle, not just the one you drew.

The three tools

EVERYTHING COMES FROM THESE

1. **Distance** $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ — for equal sides, and for Pythagoras.

2. **Midpoint** $(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2})$ — for bisection and for medians.

3. **Gradient** $\frac{y_2 - y_1}{x_2 - x_1}$ — for parallel (equal) and perpendicular (product -1).

Which shape you are proving decides which tool you reach for:

TO PROVE	SHOW THAT
parallelogram	the diagonals share a midpoint
rectangle	a parallelogram whose diagonals are also equal
rhombus	all four sides equal
right-angled triangle	Pythagoras holds, or two gradients multiply to -1
isosceles triangle	two side lengths are equal

02 Worked examples**EXAMPLE 1** Prove that $A(0; 0)$, $B(4; 0)$, $C(4; 3)$, $D(0; 3)$ is a rectangle.

① Midpoint of AC : $(2; 1.5)$

② Midpoint of BD : $(2; 1.5)$

Diagonals bisect each other $\Rightarrow$ parallelogram.

③ $AC = \sqrt{16 + 9} = 5$

④ $BD = \sqrt{16 + 9} = 5$

The diagonals are also equal.

⑤ A parallelogram with equal diagonals is a rectangle.

Theorem from Quarter I.

ANSWER

A rectangle

EXAMPLE 2 Classify the triangle $A(0; 0)$, $B(6; 0)$, $C(3; 4)$.

① $AB = 6$
Along the axis.

② $AC = \sqrt{9 + 16} = 5$

③ $BC = \sqrt{9 + 16} = 5$
Two sides equal.

④ $5^2 + 5^2 = 50 \neq 36$
Not right-angled.

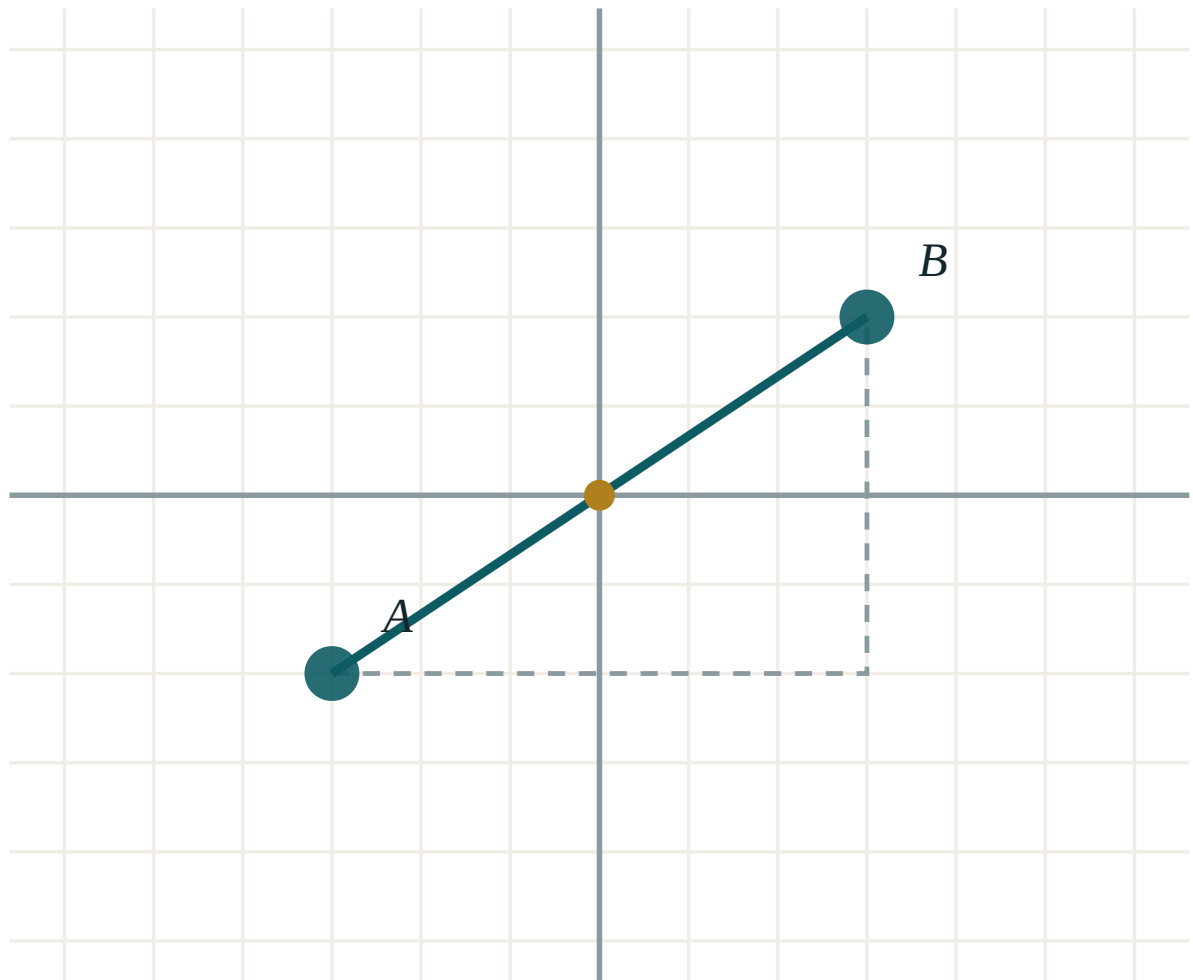
ANSWER

Isosceles, but not right-angled

03 Interactive model

Place the vertices with the model, then read off the distances the proof needs.

Measure a side of your figure



A (-3, -2)

B (3, 2)

 $x_2 - x_1$ 6 $y_2 - y_1$ 4

distance AB 7.21

midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Coordinate method	Koordinatalar usuli	Метод координат
To place on axes	O'qlarga joylashtirish	Расположить на осях
Prove	Isbotlash	Доказать
Isosceles	Teng yonli	Равнобедренный
Scalene	Turli tomonli	Разносторонний
Collinear	Bir to'g'ri chiziqda yotuvchi	Лежащие на одной прямой
Vertex	Uchi	Вершина
Diagonal	Diagonal	Диагональ
Converse	Teskari teorema	Обратная теорема

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To prove a quadrilateral is a parallelogram with coordinates, show:

all sides equal

the diagonals share a midpoint

one angle is 90°

the diagonals are equal

$(0; 0)$, $(5; 0)$, $(5; 5)$, $(0; 5)$ is a:

rectangle only

rhombus only

square

trapezium

A triangle with sides 5, 5, 6 is:

equilateral

isosceles

scalene

right-angled

Two lines are perpendicular when their gradients:

are equal

multiply to -1

add to 0

are both positive

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find the length of the side from $(0; 0)$ to $(0; 7)$.

ANSWER 7

2. Is $(0;0)$, $(4;0)$, $(4;4)$, $(0;4)$ a square?

ANSWER Yes.

3. Find the midpoint of the diagonal from $(0; 0)$ to $(6; 8)$.

ANSWER $(3; 4)$

4. Classify the triangle $(0;0)$, $(3;0)$, $(0;4)$.

ANSWER Right-angled, scalene.

5. Find the gradient from $(0; 0)$ to $(2; 6)$.

ANSWER 3

6. Are the segments from $(0;0)$ to $(2;1)$ and $(0;0)$ to $(-1;2)$ perpendicular?

ANSWER Yes — $\frac{1}{2} \cdot (-2) = -1$

7. Find the perimeter of $(0;0)$, $(3;0)$, $(3;4)$.

ANSWER 12

Medium · the standard question

7 PROBLEMS

1. Prove that $(1;1)$, $(5;1)$, $(5;4)$, $(1;4)$ is a rectangle.

ANSWER Diagonals both have midpoint $(3; 2.5)$ and both equal 5.

2. Classify the triangle $(0;0)$, $(4;0)$, $(2;5)$.

ANSWER Isosceles — $AC = BC = \sqrt{29}$

3. Show that $(0;0)$, $(4;3)$, $(8;0)$, $(4;-3)$ is a rhombus.

ANSWER All four sides equal 5.

4. Find the fourth vertex of a parallelogram with $(0;0)$, $(5;1)$, $(7;5)$.

ANSWER $(2; 4)$

5. Show that the triangle $(0;0)$, $(6;0)$, $(0;8)$ is right-angled and find its area.

ANSWER $36 + 64 = 100 \checkmark$; area 24

6. Find the length of the median from $(0;0)$ to the side joining $(6;0)$ and $(2;4)$.

ANSWER midpoint $(4; 2)$, length $\sqrt{20} = 2\sqrt{5}$

7. Are $(1;2)$, $(3;6)$ and $(6;12)$ collinear?

ANSWER Yes — both gradients equal 2, and all three satisfy $y = 2x$.

Hard · stretch and reason

7 PROBLEMS

1. Prove that the diagonals of the rectangle $(0;0)$, $(a;0)$, $(a;b)$, $(0;b)$ are equal, for all a and b .

ANSWER Both diagonals equal $\sqrt{a^2 + b^2}$.

2. Prove that the midpoints of the sides of any quadrilateral form a parallelogram.

ANSWER With vertices $(x_1; y_1) \dots (x_4; y_4)$, both diagonals of the midpoint figure have midpoint $(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4})$.

3. Show that the triangle $(1;1)$, $(4;5)$, $(8;2)$ is right-angled.

ANSWER $25 + 25 = 50$ ✓ — the right angle is at $(4; 5)$

4. Find the point on the x -axis equidistant from $(1; 2)$ and $(5; 4)$.

ANSWER $(4.5; 0)$

5. Prove that the medians of the triangle $(0;0)$, $(6;0)$, $(0;6)$ meet at $(2; 2)$.

ANSWER Each median passes through $(2; 2)$ — the centroid is the average of the vertices.

6. The vertices of a square are $(0;0)$, $(4;2)$ and $(2;6)$. Find the fourth.

ANSWER $(-2; 4)$

7. Show that $(0;0)$, $(6;0)$, $(8;3)$, $(2;3)$ is a parallelogram but not a rhombus.

ANSWER Both diagonals have midpoint $(4; 1.5)$, so it is a parallelogram; but the sides are 6 and $\sqrt{13}$, which are unequal.

07 Homework

Homework — 5 problems

Geometry 8, Tema 34. State which of the three tools you used at each step.

1. Prove that $(0;0)$, $(6;0)$, $(6;4)$, $(0;4)$ is a rectangle.
2. Classify the triangle $(0;0)$, $(8;0)$, $(4;3)$.
3. Show that $(0;0)$, $(3;4)$, $(8;4)$, $(5;0)$ is a rhombus.

4. Find the fourth vertex of a parallelogram with $(1;1)$, $(6;2)$, $(8;7)$.
5. Show that $(2;1)$, $(5;7)$ and $(8;13)$ are collinear.